An Application of Ant Colony Optimization Algorithm in TSP

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***Abstract*—This paper has made a detailed analysis of the Ant Colony Algorithm and its parameters, integrated the algorithm with the TSP problems and put forward such optimization methods as the node selection by means of piecewise function control, the pheromone updating by means of smooth elitist strategy and the adjacent edge adjustment in the global optimal solution by means of the 2-opt strategy. The mathematical model of ant colony optimization algorithm has been proposed on the basis of the analysis to solve the TSP problems and the results of simulation indicate this algorithm has a quite good performance.**

Keywords: Ant colony algorithm, TSP, node selection, smooth elitist strategy, 2-opt strategy

I. INTRODUCTION

The ant colony algorithm (ACA) comes from the research that the ants in the nature can find out the shortest path from their nest to food and back, and such algorithm has achieved good results in many industries to solve the TSP problems based on the principle as follows: the ant in movement will lay a substance called pheromone on paths, and the ant can perceive the existence of such substance and its intensity during the movement, by which the ant can be directed to move towards the direction where such substance is of higher intensity.

In the TSP problem, if n cities are given, and a traveling salesman sets out from a certain city to visit each other city once and then return to the starting city, it requires that the shortest route should be found [1]. The TSP problem is a typical NP problem and its solutions will have an exponential increase with the increase of the number of cities. In the recent years, the ant colony algorithm, the genetic algorithm, the simulated annealing algorithm and other heuristic intelligent optimization methods can effectively solve the TSP problem, among which the ant colony algorithm is widely used to solve the TSP problem due to its use of the positive feedback mechanism, heuristic probability search, distributed computing method and other characteristics.

II. ANT COLONY ALGORITHM

A. Ant system algorithm

In 1991, M.Dorigo and other Italian scientists first proposed the ant system [2] (AS).

Assume there are n cities, i, j n. m is the number of ants. The probability for Ant K moving from Node i to Node j:

$$p_{ij}^{k}(t)=\begin{cases}\dfrac{\tau_{ij}^{\alpha}(t)\cdot\eta_{ij}^{\beta}(t)}{\sum_{s\in allowed_k}\tau_{is}^{\alpha}(t)\cdot\eta_{is}^{\beta}(t)} & j\in allowed_k\\ 0 & otherwise\end{cases} \quad (1)$$

$allowed_k$ means the collection of the nodes not visited by Ant K. $\tau_{ij}(t)$ shows the amount of information along the path between City i and City j. $\eta_{ij}(t)$ is the reciprocal of the distance between i and j. α shows the relative importance of the path track. β shows the relative importance of the path visibility.

The pheromone on paths gradually disappears, ρ shows the disappearance degree of the pheromone, $0<\rho<1$, and after all ants have completed an iteration, each path will have its information updated, $\Delta\tau_{ij}(t)$ shows the increment of pheromone on ij during this iteration.

$$\tau_{ij}(t+1)=(1-\rho)\cdot\tau_{ij}(t)+\Delta\tau_{ij}(t) \quad (2)$$

B. Ant colony system algorithm

In 1996, M.Dorigo et al proposed the ant colony system [3](ACS). The improvement strategy was added based on AS:

1) Addselection strategy

Suppose to let the ant ignore the better solution when the information content has not reached a certain threshold, but have the ant approaching the path rich in accumulative information under the stimulation of such information only when the information stimulation tends to the set threshold q_0.

$$j=\begin{cases}\arg\max_{j\in allowed_k}[\tau_{ij}^{\alpha}(t)\cdot\eta_{ij}^{\beta}(t)] & q\le q_0\\ refer\ \ to(1) & otherwise\end{cases} \quad (3)$$

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DOI 10.1109/ICINIS.2012.20

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2) Add pheromone updating methods

In the t[th] iteration, when Ant K has completed a search and found a way back, then locally update the amount of information along the path k has visited. $l_{besttravel}$ indicates the optimal path the ant has passed in this iteration, i.e., the local optimal solution.

$$\tau_{ij}(t+1)=\begin{cases}(1-\rho)\cdot\tau_{ij}(t)+\Delta\tau_{ij}(t) & i,j\in l_{besttravel}\\ 0 & others\end{cases} \quad (4)$$

When the m ants all have completed their search, then globally update the amount of pheromone along the path. $g_{besttravel}$ indicates the optimal path all the ants have passed in this iteration, i.e., the global optimal solution.

$$\tau_{ij}(t+1)=\begin{cases}(1-\rho)\cdot\tau_{ij}(t)+\rho\Delta\tau_{ij}(t) & i,j\in g_{besttravel}\\ 0 & others\end{cases} \quad (5)$$

C. Maximum and minimum ant system algorithms

In 1997, T.Stutzle et al proposed the Max-Min Ant System [4] (MMAS).

1) Change pheromone update mode

This algorithm only globally updates the pheromone of the shortest path generated in this iteration. Its formula is identical to (5), but $g_{besttravel}$ in ACS is the global optimal solution in all the current iterations, while $g_{besttravel}$ in MMAS is the optimal path selected by all ants in this iteration.

2) Control pheromone amount

In order to avoid too fast convergence of the algorithm to the local optimal solution, MMAS will control the pheromone amount on each edge within a fixed range, respectively expressed by $\tau_{\min}$ and $\tau_{\max}$.

$$\tau_{\max}=\frac{1}{(1-\rho)\cdot L_{opt}} \quad (6)$$

$$\tau_{\min}=\frac{\tau_{\max}(1-\sqrt[n]{p_{best}})}{avg\cdot\sqrt[n]{p_{best}}} \quad (7)$$

L_{opt} is the optimal solution of the current problem, i.e., the shortest path; p_{best} is the probability when the pheromone of the edge the current optimal path is $\tau_{\max}$ while the pheromone of other edges is $\tau_{\min}$.

III. Optimization Strategies

A. Node selection under piecewise function control

Formula (3) shows that q_0 is a parameter to control q, and if the value of q_0 is smaller, most ants will easily select the most informative edge, so that the searched solution would have its space range restricted, which shall not be conducive to find the global optimal solution; if the value of q_0 is bigger, there will be a bigger probability for the less informative edge to be selected, and then the solution space to be searched will be enlarged, but the search presents certain blindness, and cannot be easily converged.

According to this principle, at the initial stage of the algorithm iteration, give q_0 a smaller initial value to carry out deterministic search by a higher probability, so that the edge featuring higher pheromone intensity and a shorter path would be selected to accelerate searching of the local optimal solution; however, in the middle stage of the algorithm, give q_0 a slightly bigger value to increase the random search probability and enlarge the search space; in the later stage of the algorithm, again give q_0 a smaller value to accelerate the algorithm convergence rate.

$$q_0=\begin{cases}c_1 & 0<t\le n_0\\ c_0 & n_0<t\le n_1\\ c_1 & n_1<t\le t_\max\end{cases} \quad (8)$$

t is the number of the current iterations, and t_max is the maximum number of iterations. Through the simulation test, what can be conclude is that when n_0=0.5*t_max，n_1=0.75*t_max，c0=0.5，c1=0.2, there should be the best operation results and time, and the algorithm can be effective optimized.

B. Use smooth elitist strategy to update the pheromone

The ACS global update is only to update the global optimal solution, while the MMAS global update is to update the optimal path of this iteration, and here we will have such two methods integrated with the smooth elitist strategy adopted, namely, we will aggrandize the impact of that the ant can be able to find a path better than the current optimal path on the pheromone intensity. In this way, it can be ensured that the global optimal solution could be preferably selected and the scope of solution could be effectively expanded to prevent the algorithm obtaining the local optimal solution and stagnating prematurely.

In each t[th] iteration, when one ant has its path equal to the optimal path, update the pheromone normally, in addition to this ant, the paths of other ants are bigger than this optimal path, and then use a certain percentage (< 1) to update the pheromone. At this time, the optimization shall be as follows:

$$\Delta\tau_{ij}(t)=c*Q/l_{bestlen} \quad (9)$$

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$$c=\begin{cases}1 & l_k=l_{bestlen}\\ c_k & l_k>l_{bestlen}\end{cases}\qquad 0<c_k<1 \tag{10}$$

Through analysis of the simulation test, when c_k = 0.1, the result should be optimal, and the algorithm can be effectively optimized.

C. *Use 2-opt strategy to adjust the adjacent edges in the global optimal solution*

In the search results of the ant colony algorithm, the case often appears as shown in Figure 1. For every four successive nodes of a, b, c and d in the optimal solution, rearrangement and updating of such nodes by means of the 2-opt strategy can also intensify the exploration of the solution space [5].

The 2-opt method should be as follows: For the four successive nodes of a, b, c and d in the optimal path of this iteration, when $d_{ac}+d_{bd}<d_{ab}+d_{cd}$, use the order a, c, b and d to replace the order a, b, c and d. By adjusting the node sequence, namely, exchanging the edges, make improvement within the neighborhood of some solution to optimize the global optimal solution. The effect after adjustment is as shown in Figure 2.

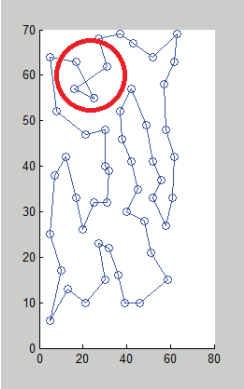


Figure 1 Shortest Path Connection

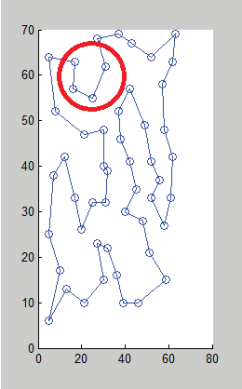


Figure 2 Shortest Path Connection after Adjustment

IV. Ant Colony Optimization Algorithm

A. *Mathematical model description*

The ant colony optimization algorithm has its mathematical model described as follows:

Step1: Suppose the TSP number is n and (x_i, y_i) is the coordinate of City i, calculate the distance (i,j n) between any two cities i and j. m is the number of ants, as for the kth ant, $allowed_k$ is the set of the nodes the ant has not visited, while $tabu_k$ is the set of the nodes the ant has visited.

Step2: When the iteration starts, and if the number of iterations is smaller than the maximum number of iterations t_max, then put the m ants randomly at n nodes, and the method for Ant k to select the next node j at the node i shall be as follows:

$$j=\begin{cases}\arg\max\limits_{j\in allowed_k}[\tau_{ij}^{\alpha}(t)\cdot\eta_{ij}^{\beta}(t)] & q\le q_0\\ \dfrac{\tau_{ij}^{\alpha}(t)\cdot\eta_{ij}^{\beta}(t)}{\sum\limits_{s\in allowed_k}\tau_{is}^{\alpha}(t)\cdot\eta_{is}^{\beta}(t)} & j\in allowed_k \quad q>q_0\\ 0 & otherwise\end{cases} \tag{11}$$

q_0 has its value based on the change of the number of iterations:

$$q_0=\begin{cases}0.2 & 0<t<0.5*t_\max\\ 0.5 & 0.5*t_\max<t<0.75*t_\max\\ 0.2 & 0.75*t_\max<t<t_\max\end{cases} \tag{12}$$

Step3: After each iteration, locally update the information amount of the path Ant k has visited:

$$\tau_{ij}(t+1)=\begin{cases}(1-\rho)\cdot\tau_{ij}(t)+\Delta\tau_{ij}(t) & i,j\in l_{besttravel}\\ 0 & others\end{cases} \tag{13}$$

$$\Delta\tau_{ij}(t)=c*Q/l_{bestlen}\qquad c=\begin{cases}1 & l_k=l_{bestlen}\\ 0.1 & l_k>l_{bestlen}\end{cases} \tag{14}$$

Step4: When all ants have finished searching, put the shortest path obtained through comparison in $g_{besttravel}$ to update the pheromone.

$$\tau_{ij}(t+1)=\begin{cases}(1-\rho)\cdot\tau_{ij}(t)+\rho\Delta\tau_{ij}(t) & i,j\in g_{besttravel}\\ 0 & others\end{cases} \tag{15}$$

Step5: Add one to the number of iterations, and return to Step2, until the maximum number of iterations has been achieved.

B. *Comparison of simulation test results*

The simulation test in this paper adopts the version of Matlab7.8.0, with three instances selected in the TSPLIB: Dantzig42, Eil76 and Ch130.

The simulation test has its parameters configured as follows: the number of ants m=2*n，Alpha=1，Beta=5，Rho=0.9，Q=100, and the simulation results incurred are as shown in Table 1.

TABLE 1 SHORTEST PATH

Instances	Best solution	Worst solution	Average solution	TSPLIB
Dantzig42	679.2	687.8	682.4	699
Eil76	544.7	564.3	557.6	538
Ch130	6284.5	8893.2	7617.3	6110

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The above results show that the ant colony optimization algorithm can effectively solve the Dantzig42 problem, and achieve better results than the TSPLIB optimal solution. According to the algorithm convergence graph, the algorithm can achieve the optimal solution in about 25 iterations through convergence, as shown in Figure 3. As for solutions of the Eil76 and Ch130 problems, the ant colony optimization algorithm also can find the path close to TSPLIB.

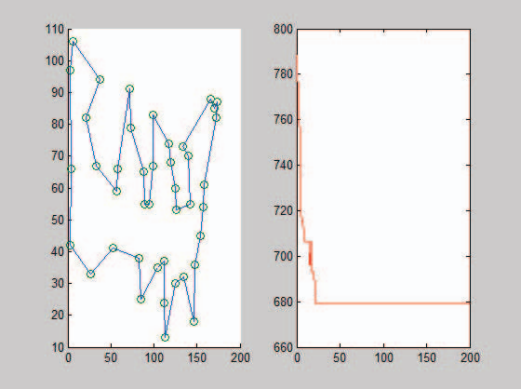


Figure 3 Solution of Dantzig42

The ant colony optimization algorithm can effectively expand the range for search of the optimal solution to prevent the algorithm obtaining the local optimal solution prematurely; at the same time, the ant colony optimization algorithm can obtain the optimal solution in a shorter period of time, which has improved the efficiency of the algorithm.

V. Conclusion

This paper proposes an ant colony optimization algorithm, by virtues of such methods as the node selection strategy of the ant colony system, the strategies for local and global pheromone update, the optimal path targeting this iteration when the max-min ant system has its pheromone globally updated, the node selection by means of piecewise function control, the pheromone update using smooth elitist strategy and the adjustment of the adjacent edges in the global optimal solution using the 2-opt strategy, to accelerate the algorithm convergence speed, expand the search space and greatly improve the performance of the improved ant colony algorithm.

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